

Assignment - 1

Software Engineering
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Food Delivery System

A. Purpose

The purpose of food delivery system is to allow customers to browse different restaurants and food items, place orders and track deliveries. It also helps manage orders and delivery requests.

B. Users

- Customers.
- Restaurant staff.
- Delivery Rider.
- System Administrator.

C. Stakeholders

- Customers
- Restaurant owner / staff
- Riders
- System Admin
- Payment Service Providers
- Food delivery company

D. Functional Requirements

→ User Registration

It would allow customers to login and create account.

→ Browse Restaurants and Menus

It would allow customers to view available food items, prices and description.

→ Place Orders

It would allow to add items to cart and place order.

→ Online Payment

It would allow customer to pay using supported payment method.

→ Order Tracking

It would allow customer to view the current state of their order

E. Non Functional Requirements

→ Performance

The system should respond to user requests within a few seconds.

→ Security

User passwords and payments info must be protected.

→ Availability

The system should remain available to users most of the time.

→ Usability

System should be simple and user-friendly.

→ Scalability

System should be able to (produce) support an increasing number of customers, restaurants and orders.

F. Risks

→ Payment Failure

Online payments may fail because of network issue.

→ Data Security Breach

Unauthorized access could expose sensitive content of customer.

→ Delivery Delays

Bad weather, traffic and incorrect address could lead to delivery delays.

G. Quality Attributes

→ Reliability

System should process orders accurately.

→ Security

System should protect customer, restaurants and payment information.

→ Usability

System should be easy to understand and operate.

H. Where AI could be used

→ Food Recommendations

AI could recommend food based on customer's previous orders and preference.

→ Delivery Time Prediction

AI could estimate delivery times using info such as traffic, distance and weather.

→ Chatbot

An AI chatbot could answer customer's common questions.

I. What could go wrong if AI is used incorrectly

→ Incorrect Recommendations

AI may recommend food that does not match customer's preference.

→ Wrong Delivery Estimates

Incorrect AI prediction could give customers unrealistic delivery times.

→ Biased or Incorrect Decision

If the AI is trained on poor or biased data, it could produce unfair recommendations or answers.